

FILTER DESIGN TEST CASE

BAW (Bulk Acoustic Wave) - Band 3 (1800 MHz) for GPS/GNSS Anti-Jamming Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a BAW (Bulk Acoustic Wave) filter targeting Band 3 (1800 MHz) for GPS/GNSS Anti-Jamming Filter. The filter is designed on AIN on Si substrate with a target center frequency of 2734.7 MHz and bandwidth of 104.1 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the GPS/GNSS Anti-Jamming Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: BAW (Bulk Acoustic Wave)
- Frequency Band: Band 3 (1800 MHz)
- Application: GPS/GNSS Anti-Jamming Filter
- Substrate: AIN on Si
- Package: Flip-Chip BGA
- Design Tool: CST Studio Suite 2024
- Design Center: Osaka Design Center

This specification applies to all variants of the BAW (Bulk Acoustic Wave) design targeting Band 3 (1800 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	2734.7 MHz
Bandwidth (3 dB):	104.1 MHz
Insertion Loss Target:	≤ 2.9 dB
Return Loss Target:	≥ 21.0 dB
Out-of-Band Rejection:	≥ 39.0 dB
Isolation (Duplexer):	≥ 36.0 dB
Q Factor:	4887
Temperature Coefficient:	-18.8 ppm/°C
Die Size:	1.4 x 1.5 mm
Wafer Size:	8-inch
Number of Resonators:	9
Electrode Material:	W

Passivation: Si3N4

4. TEST PROCEDURE

1. Open CST Studio Suite 2024 and load the BAW (Bulk Acoustic Wave) template project.
2. Define the substrate stack: AlN on Si with specified layer thicknesses.
3. Set design targets: center freq 2734.7 MHz, BW 104.1 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.9 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, AlN on Si).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 8204 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.9 dB
- Return loss in passband shall be >= 21.0 dB
- Out-of-band rejection at +/- 156 MHz offset >= 39.0 dB
- Group delay variation across passband shall be <= 10 ns
- Temperature drift over -40°C to +85°C shall be <= 6.4 MHz
- Power handling shall be >= +35 dBm at 1 dB compression
- ESD tolerance >= 1000 V (HBM)
- Die yield on qualification lot >= 90%

6. TEST RESULTS

Measurement Date: 2024-07-12
Devices Tested: 62
Insertion Loss (meas): 2.65 dB
Return Loss (meas): 23.3 dB
Rejection (meas): 42.4 dB
Isolation (meas): 40.5 dB
Q Factor (meas): 4887
Group Delay Variation: 5.6 ns
Power Handling: +35 dBm
Die Yield: 96.2%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The BAW (Bulk Acoustic Wave) design demonstrated acceptable performance for Band 3 (1800 MHz).
- b) Insertion loss is marginally above target; electrode thickness optimization recommended.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Group delay flatness improved compared to previous revision.
- e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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